ModelIt!

Student Activity Pack

The Carbon Cycle: Where Does Your Breath Go?

How Carbon Travels Between Air, Plants, Soil, and Back Again

Grade 05L2 | NGSS 5-LS1-1, 5-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 When you breathe out, where does that air go? What happens to it?

2 Do plants breathe? If so, what do they breathe in and out?

3 Draw arrows showing how carbon might move between a tree, the air, and the soil.

4 What do you think happens to a fallen leaf over time?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- CO2 in Atmosphere
- Sunlight Intensity
- Photosynthesis Rate
- Decomposition Rate
- Soil Carbon Storage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Carbon Cycle

Moderate CO_2 and sunlight levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sunlight Shutdown

Lock Sunlight Intensity to 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High CO_2 World

Set CO_2 to 90%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ocean Absorption — Oceans absorb CO_2 from the atmosphere

Fossil Fuel Burning — Humans release stored carbon by burning coal and oil

Animal Respiration — Animals breathe out CO_2 from digesting food

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Where does the carbon in your breath go, and how does it come back? Use evidence from your simulation to support your answers.

1. **Why is the carbon cycle called a cycle and not a chain?**

2. **What would happen to CO_2 levels if all decomposition suddenly stopped?**

3. **How are forests sometimes called 'the lungs of the Earth'?**

4. **If you could add one more component to this model, what would it be and why?**

5. **How does understanding the carbon cycle help us think about climate change?**

8 STEM Challenge: Design a Carbon-Capturing Schoolyard

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YOUR ENGINEERING MISSION

Design a schoolyard landscape that maximizes carbon capture from the atmosphere.

THE SCENARIO

Your school district wants to reduce its carbon footprint. They've asked your class to redesign the schoolyard to absorb as much CO_2 as possible.

Engineering Design Thinking

1. WHAT MIX OF TREES, SHRUBS, AND GROUND COVER WILL YOU PLANT?
2. HOW WILL YOU MAXIMIZE SUNLIGHT EXPOSURE FOR YOUR PLANTS?
3. HOW WILL YOU MANAGE FALLEN LEAVES AND ORGANIC MATTER?
4. HOW WILL YOU MEASURE WHETHER YOUR DESIGN IS ACTUALLY CAPTURING CARBON?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists and Carbon Analysts study how carbon moves through Earth's systems and develop strategies to reduce CO_2 in the atmosphere. They earn $70,000-$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the carbon cycle model, what happens when Sunlight Intensity is locked at 0%?

- **A.** Photosynthesis Rate increases because plants use stored energy
- **B.** CO_2 in Atmosphere decreases because decomposition slows down
- **C.** CO_2 in Atmosphere rises because photosynthesis stops but decomposition continues releasing CO_2
- **D.** Soil Carbon Storage increases because plants grow underground

2 What type of relationship exists between Photosynthesis Rate and CO_2 in Atmosphere?

- **A.** Positive: more photosynthesis adds more CO_2 to the air
- **B.** Negative: more photosynthesis removes CO_2 from the air
- **C.** No relationship: photosynthesis does not affect CO_2
- **D.** Positive: photosynthesis and CO_2 always increase together

3 In the model, when CO_2 in Atmosphere is set to 90%, what happens to Photosynthesis Rate and why?

- **A.** Photosynthesis Rate drops because too much CO_2 poisons plants
- **B.** Photosynthesis Rate increases because plants have more CO_2 raw material to use
- **C.** Photosynthesis Rate stays the same because plants only need sunlight
- **D.** Photosynthesis Rate drops to zero because CO_2 blocks sunlight

4 Why is the carbon cycle called a 'cycle' and not a 'chain'?

- **A.** Because carbon is destroyed at the end of the process
- **B.** Because carbon moves in only one direction, from air to soil
- **C.** Because carbon atoms loop continuously between the atmosphere, plants, soil, and back again
- **D.** Because the cycle only happens once per year

5 A scientist notices that a forest has very little dead organic matter on the ground despite many trees dropping leaves. What does this tell you about the carbon cycle in this forest?

- **A.** The trees have stopped producing leaves
- **B.** Decomposition Rate is very high, breaking down dead matter quickly and releasing carbon back to the atmosphere
- **C.** The wind blows all the leaves to another forest
- **D.** Soil Carbon Storage has reached zero and can't hold any more